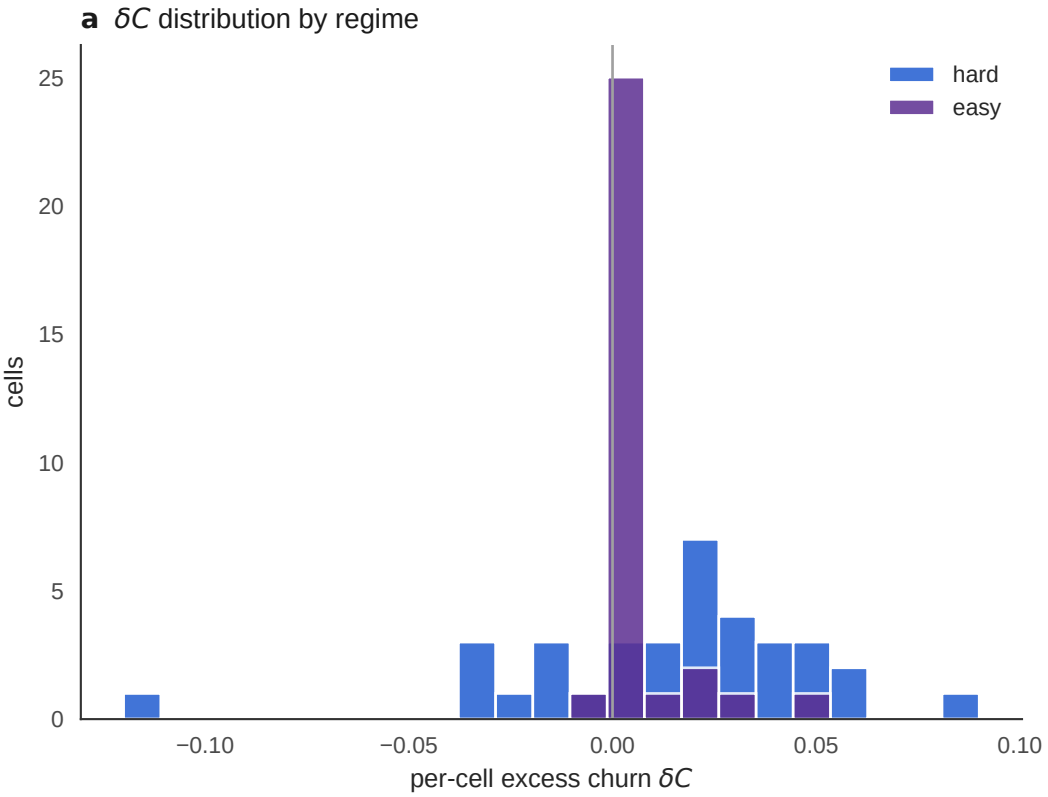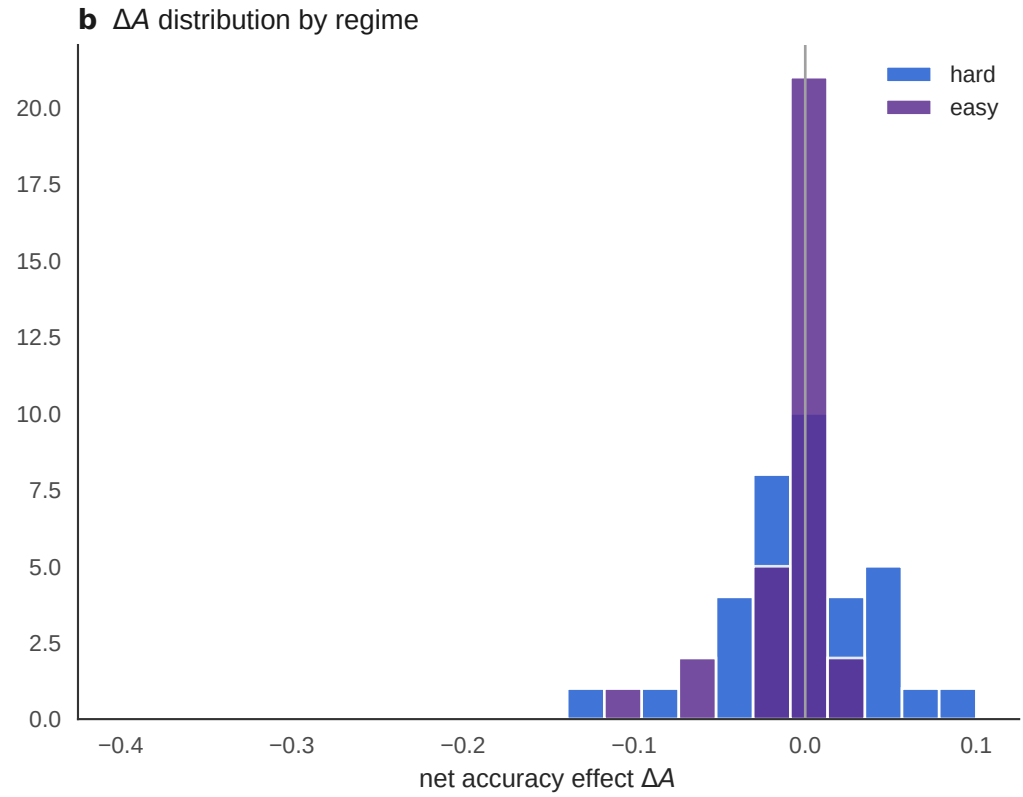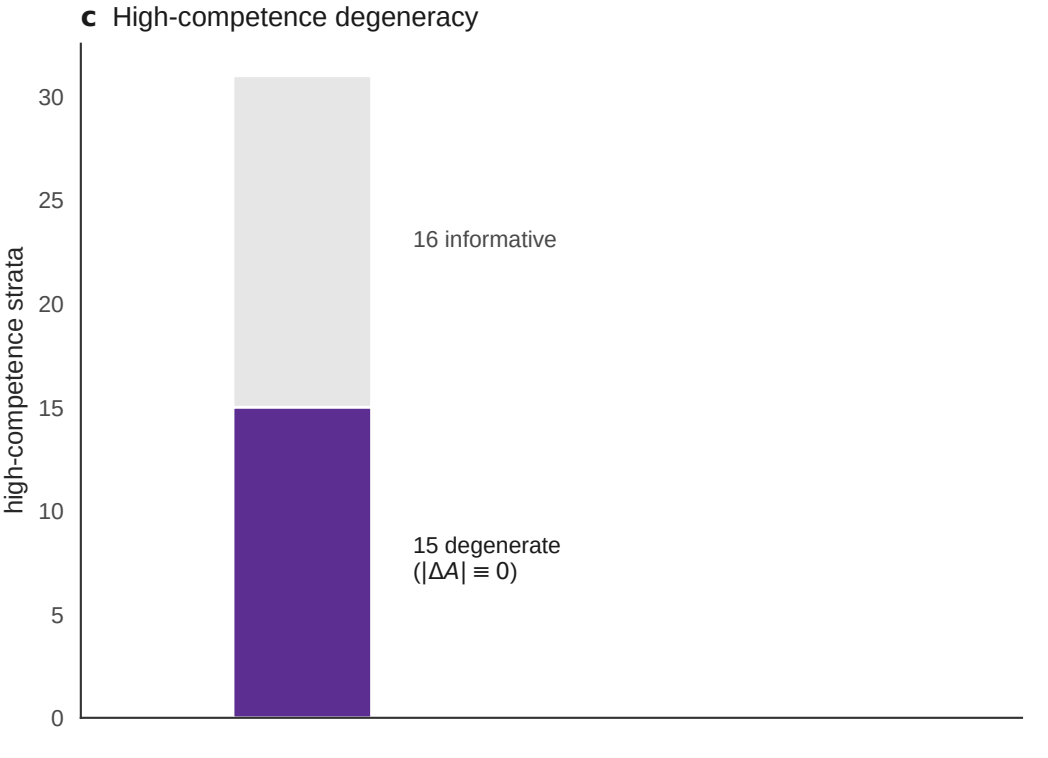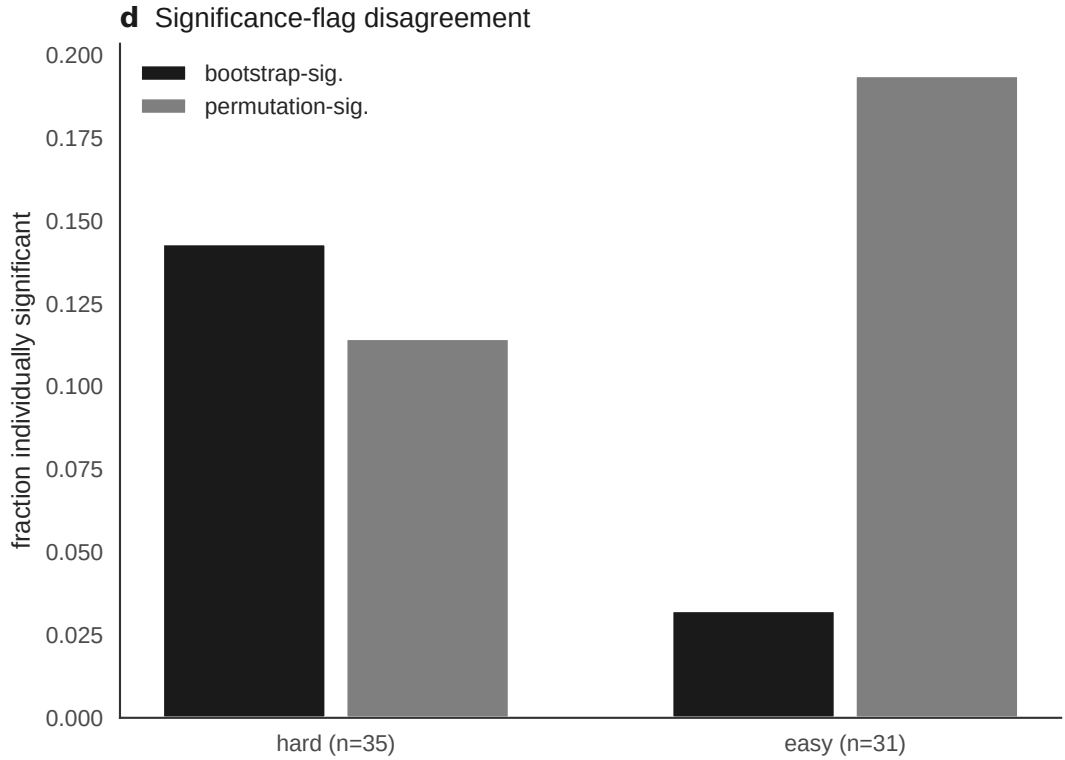Easy / hard separation atlas